

# Segurança em Redes de Comunicações

## Security in Communications Networks

### First Project

Professors:

|                |                                                              |
|----------------|--------------------------------------------------------------|
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**Objective:** Plan, deploy and perform operational tests of flow control policies in a corporate network.

#### Description:

- Consider the (simplified) network in appendix. Router/Firewall devices are only configured as routers, no security policies have been deployed. Basic configurations will be provided.
- Use the network 10.0.0.0/16 to the internal connections, the network 10.1.0.0/24 for VLAN1, the network 10.10.0.0/24 for VLAN10, the network 10.20.0.0/24 for VLAN20, 10.100.0.0/16 for the internal Datacenter, and network 200.0.0.0/24 for the DMZ. You may use network 100.0.0.0/24 for testing Internet services.
- Consider that the network have the following services (each one deployed in more than one server):
  - ♦ In the DMZ: Web services (HTTPS UDP/TCP 443), E-mail (IMAP TCP 993 and SMTP TCP 25), DNS (UDP 53).
  - ♦ In the internal Datacenter: Intranet/Storage (TCP 443), internal DNS (DNS 53), Databases (TCP 3306).
- **Note: You may use a single server/terminal in each network to deploy/simulate all services.**
- Consider the following security policies
  1. The network should be able to handle high-rate DDoS attacks from the Internet.
  2. Internal devices may only access Internet services using ports TCP/UDP 80 and 443;
  3. VLAN10 devices may access VoIP services using ports UDP 5060, but only for a set of predefined IPv4 addresses;
  4. All DMZ services must be accessible from the Internet and from inside the network;
  5. The intranet/storage and internal DNS may only be accessible by devices on VLAN10 and 20.
  6. The internal Databases may only be accessible by devices on VLAN 20.
  7. A specific device on VLAN 1 (Building A) can PING and use SSH (port TCP 22) to access the console of all network devices.
  8. VLAN 10 devices may only use File Sharing (Samba, TCP ports 139 and 445) to communicate directly with VLAN 20 devices, and vice-versa.
- Define the placement of any additional firewalls, load-balancers and zones required to implement the security policies.
- **Note2: For simplification, you can reuse the core Routers (C1 and C2) as load-balancers.**
- Deploy the network using GNS3, and implement and test the security policies on the deployed network.
- A demonstration of the deployment and operational tests of the scenario described above should be done on the classes of May 5<sup>th</sup> week.

Should be done by a group of 2 students. Exceptionally, can be done individually.
- Task grading:
  - Network routing and connectivity (3 points).
  - High-availability scenarios configuration (6 points).
  - Zones definition [based on a security policies] (4 points).
  - Inter-zone rules [based on a security policies] (7 points).
  - Demonstration (per-person multiplication factor: 0% to 110%).

